

Falsification probes separate shared response agreement from perturbation-specific retrieval

A Response agreement

Audit-delta Pearson

K562

RPE1

Mean-effect

0.387

0.635

Label-shuffled

0.153

0.387

GEARS

0.284

0.462

0.0 0.2 0.4 0.6

Audit-delta Pearson

0.0 0.2 0.4 0.6

Audit-delta Pearson

B Perturbation-specific retrieval

Mean reciprocal rank (MRR)

K562

RPE1

0.027

0.020

0.033

0.019

0.050

0.026

0.00 0.02 0.04 0.06

MRR

0.00 0.02 0.04 0.06

MRR

○ Mean-effect: target-blind response probe/baseline

△ Label-shuffled: target labels scrambled diagnostic probe

● GEARS: target-aware predictive model

Probe survival narrows the interpretation of response agreement